

AI-powered bioinformatic analysis using R, #1

Course Overview & AI-assisted Setup

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About This Course

- **“AI-powered bioinformatic analysis using R”**
- HUST Bioinformatics course series (16 lectures)
- **Part 1** (Talks 1–6): R programming foundations
- **Part 2** (Talks 7–16): multi-omics data analysis
- **AI-assisted approach:** CodeBuddy throughout the entire course

Course Structure — Part 1

Talk	Topic
1	Course Overview & AI-assisted Setup ← today
2	R Language Basics I — data types, vectors, matrices
3	R Language Basics II — factors, lists, data frames
4	Data Wrangling with tidyverse (dplyr + tidyr)
5	Data Visualization with ggplot2
6	Strings, Iteration & Functional Programming

Course Structure — Part 2

Talk	Topic
7	Statistical Analysis & Machine Learning Basics
8	Regression & Survival Analysis
9	Transcriptome Analysis (DEGs, functional enrichment)
10	Microbiome Analysis (16S / metagenomics)
11	Genome Analysis (GWAS)
12	Single-cell RNA-seq Analysis
13	Spatial Transcriptomics
14	Network Analysis (WGCNA, PPI, GRN)
15	Multi-omics Integration & Case Studies
16	Other Omics & Course Wrap-up

Target Audience

- Undergraduate students in bioinformatics, computational biology, statistics, data science, or biology
- **No prior programming experience required**
- Basic biology knowledge helpful but not mandatory

Learning Approach

- **AI-assisted:** CodeBuddy prompt-driven throughout
- **Hands-on:** live coding in every session
- **Exercises:** Quarto (.qmd) format after key lectures
- **Build a portfolio** of reproducible analysis scripts

A Brief History of R

- Created in **1993** by **Ross Ihaka** and **Robert Gentleman**
 - University of Auckland, New Zealand
- Name: first letters of the two creators' first names
 - Also a playful reference to the **S language** (Bell Labs)
- **1993–2000**: small but growing academic community
- **2000**: R 1.0.0 released — the tipping point
- Now maintained by the **R Core Team** (~20 experts worldwide)
- Latest stable version: R 4.x (annual releases)

Why R Succeeded

External factors:

- Rise of free / open-source software movement
- Linux ecosystem maturity in the 2000s
- Economic pressures (free alternative to SAS, SPSS, Stata)

Internal strengths:

- Professional-grade statistical computing
- Unmatched data visualization (ggplot2)
- Rich package ecosystem: **CRAN** (>20,000 packages) + **Bioconductor**
- Strong, welcoming community
- Excellent IDE: RStudio (now Posit)

R's Popularity Today

- Consistently **top-10** in TIOBE and PYPL programming language rankings
- The **most used** statistical software in academic publications
- Increasing adoption in pharma, biotech, finance, tech
- Growing demand in the job market (Indeed.com trends)

Worldwide, Aug 2025 :

Rank	Change	Language	Share	1-year trend
1		Python	30.5 %	+0.9 %
2		Java	15.54 %	+0.2 %
3	↑↑	C/C++	8.3 %	+1.8 %
4	↓	JavaScript	7.32 %	-1.0 %
5	↓	C#	5.32 %	-1.3 %
6		R	5.19 %	+0.5 %
7	↑↑↑↑	Objective-C	3.57 %	+1.2 %

R for Bioinformatics

Why R is an excellent choice for bioinformatics:

- **Bioconductor**: 2,000+ bioinformatics packages (RNA-seq, ChIP-seq, flow cytometry, etc.)
- **tidyverse**: elegant, consistent data wrangling & visualization
- **Reproducible research**: Quarto / R Markdown for literate programming
- Strong **genomics & statistics** community
- **Interoperability**: R-Python via reticulate, C++ via Rcpp, shell commands



Tip

R + CodeBuddy AI = the most productive setup for bioinformatics today.

What we will do today

- Set up an **AI-assisted programming** environment
- Use **CodeBuddy** as your AI coding companion
- Accomplish everything via natural language (**prompts**)
- Install R / RStudio / tinytex / tidyverse
- Clone the course repo and open the R project

Traditional vs. AI-assisted

Traditional	AI-assisted
Search for installers	AI installs directly
Follow tutorials	One sentence, done
Memorize commands	Describe what you want
Debug alone	AI helps diagnose

What is CodeBuddy?

- **Tencent Cloud** AI-native IDE (Integrated Development Environment)
- Powered by **Tencent Hunyuan + DeepSeek** dual models
- Supports **200+ programming languages** (R, Python, Julia, etc.)
- Three product forms:
 - **Standalone IDE** (what we use in this course)
 - IDE Plugins (VS Code, JetBrains)
 - CLI command-line tool

Why CodeBuddy?

- Free to use, login with **WeChat QR scan**
- **Agent mode**: AI autonomously completes multi-file tasks
- Deep integration: terminal, Git, extension marketplace
- Full R / Quarto / LaTeX pipeline support
- Native Chinese IDE, no VPN needed

Download

- ❶ Open your browser and visit: **<https://www.codebuddy.cn>**
- ❷ Click **Download**, choose your system:
 - **macOS Apple Silicon** (M1/M2/M3/M4 chip)
 - **macOS Intel** (Intel chip)
 - **Windows**
- ❸ To find your Mac chip type: Apple menu → “About This Mac” → look at the “Chip” field

Install

- **macOS:** drag the installer into the Applications folder
- **Windows:** double-click the installer and follow the wizard



Tip

After installation, open CodeBuddy from your Applications / Start menu.

Sign in

- 1 Open CodeBuddy IDE
- 2 Click the **Login** button (top-right corner)
- 3 Select **WeChat** login
- 4 Scan the QR code with your phone → confirm
- 5 You are now signed in

! Important

On first launch, you may be asked to choose the UI language. We recommend **English** for consistency with this course.

Core interface layout

Panel	Location	Function
File Explorer	Left sidebar	Browse / manage project files
Editor	Center	View and edit code
Terminal	Bottom	Execute command-line operations
AI Chat	Right sidebar (Cmd+Ctrl+I)	Interact with AI

Agent modes

CodeBuddy offers three core **Agent modes**:

- **Ask** — Q&A only; does not modify code
- **Craft** — Code generation and local edits (agent mode)
- **Plan** — AI drafts a plan first, then executes after your confirmation



Tip

Switch modes at the bottom of the chat panel, or use **Cmd + I** (macOS) as the shortcut.

Activate Craft Agent mode

This course uses **Craft (Agent) mode** throughout — let AI complete tasks autonomously:

- 1 Open the AI chat panel (Cmd + Ctrl + I)
- 2 Switch to **Craft** at the bottom of the panel
- 3 Type your **prompt** into the input box

What is a Prompt?

- A **prompt** is the natural-language instruction you give to AI
- No need to memorize commands — just describe what you want
- Example: “Please help me install R 4.4 on my Mac.”

! Important

From this point forward, **every operation** in this course will be driven by prompts in CodeBuddy.

Install R

Type the following prompt in CodeBuddy's Craft mode:

Prompt

```
Please help me install the latest version of R on my computer.  
My OS is macOS / Windows.  
If R is already installed, check the version and let me know.
```


What CodeBuddy will do

CodeBuddy will:

- 1 Detect your operating system
- 2 Guide you to download R from CRAN
- 3 Or install directly via **Homebrew** (macOS) / **winget** (Windows)



Tip

To verify: open the terminal and type `R --version`

Install RStudio Desktop

Prompt

Please help me install the latest RStudio Desktop (free edition).
If it is already installed, skip this step.
Use the terminal or provide a download link.

What CodeBuddy will do

CodeBuddy will:

- 1 Visit the Posit website for the latest RStudio download link
- 2 Guide the installation or install via package manager



Tip

After installation, RStudio will appear in your Applications (macOS) or Start menu (Windows).

R-related extensions

Install these extensions in CodeBuddy for optimal R programming:

Extension	Purpose
R Extension (<code>REditorSupport.r</code>)	R syntax highlighting, code completion, run R code
R Debugger (<code>RDebugger.r-debugger</code>)	R code debugging
Quarto (<code>quarto.quarto</code>)	.qmd file editing, preview, and rendering



Install extensions via AI

i Prompt

Please help me install these extensions in CodeBuddy:

1. R Extension (REditorSupport.r) — R syntax highlighting and code execution
2. R Debugger (RDebugger.r-debugger) — R debugging support
3. Quarto (quarto.quarto) — QMD file editing and preview

Search and install them from the extension marketplace.



Clone from GitHub

All course materials are hosted at:

<https://github.com/evolgeniusteam/R-for-bioinformatics>

Prompt

Please clone the following GitHub repository to my Desktop:
`https://github.com/evolgeniusteam/R-for-bioinformatics`

Use `git clone` to download it to `~/Desktop/`.

Open in CodeBuddy

Prompt

Please open the following folder in the current CodeBuddy window:
`~/Desktop/R-for-bioinformatics`

Open the R Project in RStudio

The repository contains an .Rproj file for RStudio:

Prompt

```
Please help me open this R project file in RStudio:  
~/Desktop/R-for-bioinformatics/R for bioinformatics.Rproj
```

```
If RStudio is not in PATH, locate it and launch it.
```


CodeBuddy vs. RStudio workflow

Task	Where
AI-assisted coding & debugging	CodeBuddy
Traditional R scripting	RStudio
Rendering QMD to PDF	Either (via terminal or RStudio)
Git operations	CodeBuddy terminal or integrated Git



Tip

You can freely switch between CodeBuddy and RStudio throughout the course — both work with the same project folder.

Install tinytex (LaTeX engine)

PDF output requires LaTeX; we recommend **tinytex**:

Prompt

Please help me install the tinytex R package.
After installation, run `tinytex::install_tinytex()` to install TinyTeX.

CodeBuddy will execute in the R terminal:

```
install.packages("tinytex")  
tinytex::install_tinytex()
```



Course R packages

Prompt

Please help me install these R packages:

- rmarkdown — R Markdown support
- knitr — code chunk execution engine
- bookdown — long document support
- kableExtra — table formatting

If compilation is needed, install from binary (type = "binary").

What is tidyverse?

tidyverse is a **meta-package** containing the core tools for data analysis:

Package	Purpose
ggplot2	Data visualization
dplyr	Data manipulation
tidyr	Data tidying
readr	Data import
purrr	Functional programming
tibble	Modern data frames
stringr	String handling
forcats	Factor variables

Install via AI

Prompt

Please install the tidyverse package in R. tidyverse is a large collection and may take a few minutes. Use `install.packages("tidyverse")` and wait. After installation, run `library(tidyverse)` to verify.

Verify installation

A successful installation will show:

```
library(tidyverse)
#>   Attaching core tidyverse packages      tidyverse 2.0.0
#>   dplyr      1.1.4      readr      2.1.5
#>   forcats    1.0.0      stringr    1.5.1
#>   ggplot2    3.5.1      tibble     3.2.1
#>   lubridate  1.9.3      tidyr      1.3.1
#>   purrr      1.0.2
```

Part 1

Please confirm these items are complete:

- ☐ CodeBuddy IDE installed and signed in with WeChat
- ☐ Craft Agent mode activated
- ☐ R installed (`R --version` in terminal)
- ☐ RStudio installed
- ☐ R / Quarto extensions installed in CodeBuddy

Part 2

- ☐ Course repository cloned to local machine
- ☐ R Project can be opened in RStudio
- ☐ tinytex installed (`tinytex::is_tinytex()` returns TRUE)
- ☐ rmarkdown / knitr / bookdown / kableExtra installed
- ☐ **tidyverse installed and loads without errors**

! Important

If any item above is unchecked, ask CodeBuddy AI to help you fix it!

Verification 1 — Beamer PDF with user info

Use CodeBuddy to create the QMD file, then **manually render** it in RStudio:

Prompt

Please create a new QMD file at the project root called "verification-beamer.qmd". Use beamer format (1610 aspectratio, Warsaw theme, xelatex engine).

The PDF should contain:

1. A title slide: "Verification Report — Beamer"
2. A slide listing my system info: user name, R version, platform (OS)
3. A slide showing my current working directory (getwd())
4. A slide listing all installed packages, grouped by their library path

Include R code chunks (echo=TRUE) for each item so I can see the code and output together.



Verification 1 — Beamer PDF with user info, cont.

Expected result

After CodeBuddy creates the file:

- 1 Open `verification-beamer.qmd` in **RStudio**
- 2 Click the **Render** button (or press `Cmd+Shift+K` on macOS / `Ctrl+Shift+K` on Windows)
- 3 RStudio will generate `verification-beamer.pdf`

A `verification-beamer.pdf` appears in the project root. Open it to verify all information is correctly displayed.

Verification 2 — Homework-style PDF with user info

Use CodeBuddy to create the QMD file, then **manually render** it in RStudio — this simulates exactly how you will submit future homework:

Prompt

Please create a new QMD file at the project root called "verification-homework.qmd". Use article format (output: pdf_document) with these R code chunks:

1. A top-level heading: "Verification Report — Homework"
2. My user name: `Sys.info()["user"]`
3. My R version: `R.version.string`
4. My operating system: `Sys.info()["sysname"]`, `Sys.info()["release"]`
5. My current working directory: `getwd()`
6. A data frame (using tibble) listing all installed packages,
columns: Package, Version, LibPath

Set `echo=TRUE` on all chunks and add brief text explanations before each chunk.



Verification 2 — Homework-style PDF with user info, cont.

Expected result

After CodeBuddy creates the file:

- 1 Open `verification-homework.qmd` in **RStudio**
- 2 Click the **Render** button (or press `Cmd+Shift+K` on macOS / `Ctrl+Shift+K` on Windows)
- 3 RStudio will generate `verification-homework.pdf` — just like real homework!

A `verification-homework.pdf` appears in the project root. This is exactly the workflow you will use for every homework submission in this course.

Verification 1+2 — Checklist

- ☐ verification-beamer.pdf generated and displays correctly
- ☐ verification-homework.pdf generated and displays correctly
- ☐ Both PDFs show correct: user name, R version, working directory, package list

! Important

If either PDF fails to render, ask CodeBuddy to debug the error — this is excellent practice for the real homework workflow!

Best practices

“Clear input → better output”

- **State the goal** — what you want to do (install, run, debug)
- **Describe the environment** — OS, software versions
- **Describe the expected outcome** — what success looks like
- **When errors occur** — paste the error message directly to AI

Prompt templates

Scenario	Prompt
Install software	"Please install [name] version [X.X] on [OS]"
Run code	"Run this R code and explain the results: [code]"
Debug error	"I got this error: [message]. Please fix it."
Learn a concept	"Explain [R concept] in simple terms with examples"



After today

- ① **Explore CodeBuddy:** practice using AI chat, get comfortable with prompts
- ② **Browse the course repo:** check out data/, rscripts/, Exercises and homework/
- ③ **Test tidyverse:** load it in R, verify everything works
- ④ **Preview next class:** R language basics

Reference Links

- CodeBuddy: <https://www.codebuddy.cn>
- CodeBuddy Docs: <https://www.codebuddy.ai/docs>
- R Project: <https://www.r-project.org>
- RStudio: <https://posit.co/products/open-source/rstudio/>
- Course Repository: <https://github.com/evolgeniusteam/R-for-bioinformatics>
- tidyverse: <https://www.tidyverse.org>
- R for Data Science (2e): <https://r4ds.hadley.nz> (must-read!)